H9DAI
Data Analytics for Artificial Intelligence
Lab 1

PGDAI

Semester 2: 2026

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Week-1-Lab_1 Manual: EDA

Instructions:

- **Use any of the Jupyter Notebooks, Google Colab, Anaconda or whatever you are familiar with for using Python**
- **Follow "hello_world_eda v2.ipynb" using *titanic* dataset, extend it further with your own code snippets for EDA.**

Once you get familiar practice EDA yourself using

1. Datasets: Download anyone of dataset from the list below

- **Students Performance:**
  https://www.kaggle.com/datasets/spscientist/students-performance-in-exams

- **Students Test Data:**
  https://www.kaggle.com/datasets/atharvbharaskar/students-test-data

- **Credit Card Fraud Detection Dataset 2023:**
  https://www.kaggle.com/datasets/nelgiriyewithana/credit-card-fraud-detection-dataset-2023

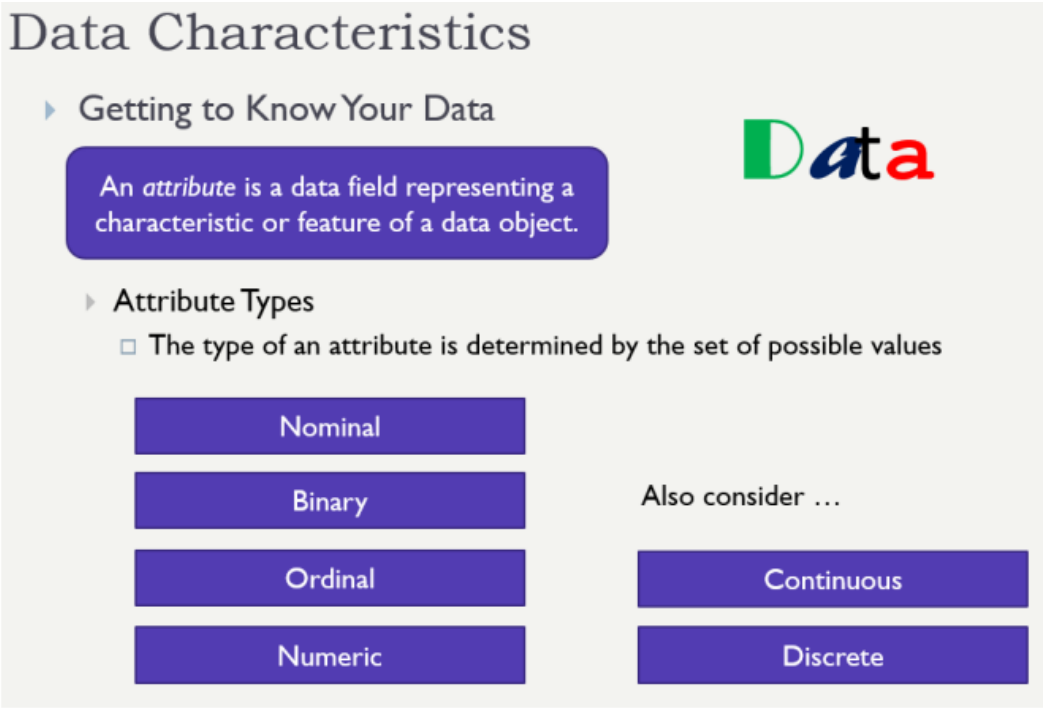
- **Salary Dataset Of Business Levels:**
  https://www.kaggle.com/datasets/taseermehboob9/salary-dataset-of-business-levels

- **World University Rankings 2023:**
  https://www.kaggle.com/datasets/samiatisha/world-university-rankings-2023-clean-dataset

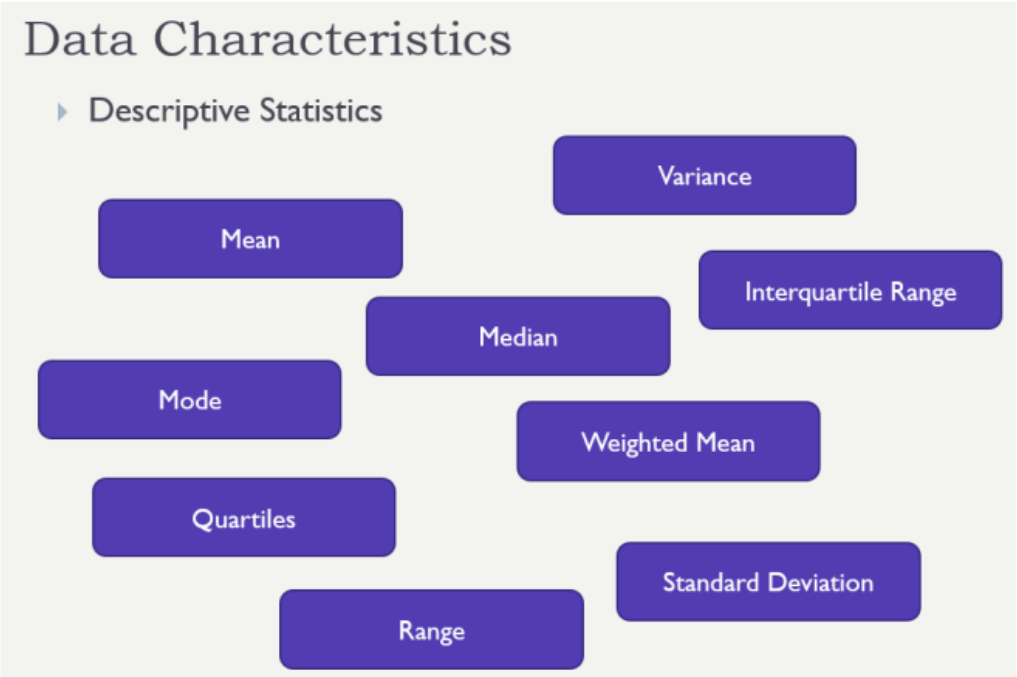
Further sources of data can also be explored such as UCI Machine Learning Repository, Data.gov, OpenML, AWS Public datasets etc.

2. Tasks: Analyse the data characteristics in the datasets above

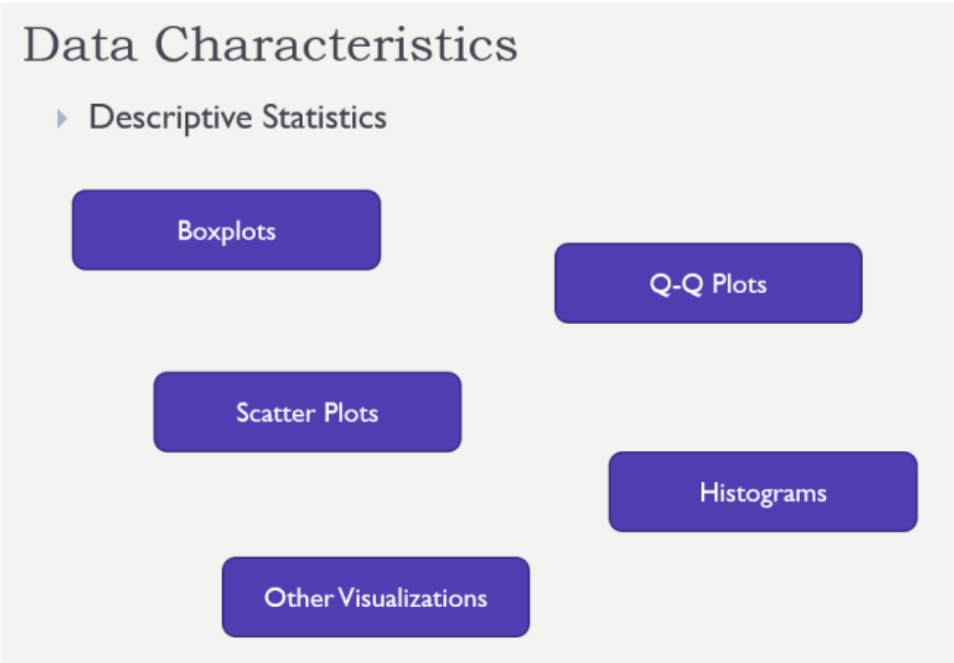
Task 1: Identify attribute types in the data



Task 2: Analyse descriptive characteristics of the data



Task 3: Visualise data characteristics



Task 4: Analyse the data

- **Missing values**
- **Distortions**
- **Mis-recordings**
- **Inadequate sampling**
- **Outliers**